

School of Computer Science and Engineering

Mini Project – One page Write up

VI Semester B.Tech – AI & DS and B.Tech – CSE (AI & ML) – 2026

Date:13-03-2026

Title: Adaptive Pathfinding on Hexagonal Grids using Reinforcement Learning Assisted A*

DOMAIN: Artificial Intelligence / Reinforcement Learning / Intelligent Systems

Problem Statement:

Pathfinding algorithms are widely used in robotics, autonomous vehicles, logistics systems, and game AI to determine optimal routes between locations. Traditional algorithms such as the A* search algorithm are efficient in static environments but often struggle when dealing with dynamic obstacles, moving goals, or partially unknown environments.

Reinforcement Learning (RL) provides an alternative approach where an agent learns optimal navigation strategies by interacting with the environment. However, RL alone can require large training time and computational resources. A hybrid approach that combines classical pathfinding algorithms with reinforcement learning can potentially improve adaptability and efficiency.

This project aims to develop an intelligent pathfinding system on hexagonal grid environments that integrates Reinforcement Learning with the A* algorithm to learn adaptive heuristics and improve navigation in dynamic environments.

Objectives:

- To design a hexagonal grid-based pathfinding environment for navigation simulation.
- To implement the A* algorithm as the baseline pathfinding method.
- To develop a Reinforcement Learning model (Q-Learning or Deep Q Network) for navigation.
- To use RL to learn adaptive heuristics that improve A* search performance.
- To evaluate pathfinding performance in static, dynamic, and partially unknown environments.
- To analyze metrics such as path optimality, computation time, nodes explored, and success rate.

Expected Outcome:

- A simulation framework for hexagonal grid pathfinding and visualization.
- Implementation of A, Reinforcement Learning, and Hybrid RL-assisted A* algorithms.
- An adaptive heuristic model that improves pathfinding efficiency.
- Performance comparison of different algorithms under multiple environmental conditions.
- A prototype system demonstrating intelligent and adaptive navigation strategies.

School of Computer Science and Engineering

Mini Project – One page Write up

Technology Stack:

- **HTML5** – structure of the web interface
- **CSS3 / Tailwind CSS** – styling and layout
- **JavaScript** – client-side logic
- **React.js** – interactive UI and visualization components
- **D3.js / Three.js / Chart.js** – visualization of hexagonal grid, paths, and performance metrics
- **Python** – core programming language
- **Flask / FastAPI** – backend API framework
- **NumPy / SciPy** – numerical computations
- **PyTorch / TensorFlow** – implementing Reinforcement Learning models (Q-learning / DQN)
- **NetworkX / custom algorithms** – A* and graph-based pathfinding implementation
- **Matplotlib / Plotly** – path and performance visualization
- **Gymnasium / custom RL environment** – training RL agents on hex grids

Methodology to solve the Problem Statement:

1. Design a hexagonal grid environment representing navigation space with obstacles and goal states.
2. Implement the *A algorithm using hexagonal distance heuristics** as the baseline pathfinding approach.
3. Develop a Reinforcement Learning agent using Q-learning or Deep Q Networks to learn navigation strategies.
4. Integrate RL with A* to learn adaptive heuristics that guide the search process.
5. Create different test scenarios including static obstacles, dynamic obstacles, moving goals, and partially unknown maps.
6. Evaluate the algorithms using metrics such as path length, computation time, number of explored nodes, success rate, and energy cost of paths.
7. Visualize and analyze the results to identify the most efficient pathfinding approach.

Team Details:

R23EH115 - Saqib Ahmed K -

R23EH075 - Mohammed Umar -

Guide Signature